

ESP Troubleshooting Guide



UNBRIDLED ESP Systems

CONDITION

PARAMETER

	Flow Rate	Wellhead Pressure	Operating Amps	Pump Discharge Pressure	Pump Intake Pressure	Motor Temperature
Broken Shaft	↘ Reduced or No Flow	↘ Low or Facility Pressure Only	↘ Lower or Idle Motor Amps	↘ Lower or equal to Intake Pressure	↗ Increasing	↗ Increasing or High Temp Trips
Hole in Tubing	↘ Reduced or No Flow	↘ Low or Facility Pressure Only	↗ Steady or Increasing	↘ Decreasing	↗ Increasing	↗ Increasing or High Temp Trips
Blockage at Pump Intake	↘ Reduced or No Flow	↘ Low or Facility Pressure Only	↘ Lower or Idle Motor Amps	↘ Lower or equal to Intake Pressure	↗ Increasing	↗ Increasing or High Temp Trips
Blockage at Perforations	↘ Reduced Flow	↘ Decreasing	↘ Decreasing	↗ Steady or Increasing	↗ Steady or Decreasing	↗ Increasing
Start-up with Kill Fluid	↗ Steady or Increasing	↘ Reduced	↗ Elevated	↗ Elevated	→ Steady	↗ Elevated
Shut in at Surface	↓ No Flow	↑ Excessive	↘ Lower or Idle Motor Amps	↗ Elevated	↗ Increasing	↑ Increasing or High Temp Trips
Blockage in Pump Stage	↘ Reduced or No Flow	↘ Reduced	↘ Reduced	↘ Reduced	↗ Increasing	↗ Increasing or High Temp Trips
Increase in Reservoir Pressure	↗ Steady or Increasing	↗ Elevated	↗ Elevated	↗ Steady or Reduced	↗ Increasing	↗ Steady or Reduced
Increase of Free Gas at Pump Intake	↘ Reduced or No Flow	↘ Reduced	↕ Interic	↕ Driftic	↗ Increasing	↗ Increasing or High Temp Trips
Stage Wear	↘ Reduced Flow	↘ Decreasing	↗ Elevated	↘ Reduced	↗ Elevated	↗ Increasing
Increase in Frequency	↗ Increasing	↗ Elevated	↗ Elevated	↗ Elevated	↘ Reduced	↗ Elevated